

SIMATS ENGINEERING

Assessment Tool 1: Short Answer Test — Solutions

Course Name	Cloud Computing and Big Data Analytics
Course Code	CSA1158
Course Outcome	CO1: Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)
Name / Reg. No.	Sooraj T 192524120

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Signature of the student: _____

Q1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.

Definition

Cloud computing is a model for delivering computing resources — servers, storage, databases, networking, and software — on demand over the internet. Instead of an organization buying, installing, and maintaining its own physical hardware, a cloud provider owns and operates the infrastructure, and customers access it remotely and pay only for what they use.

Four Distinguishing Characteristics

- **On-Demand Self-Service** — Users can provision servers, storage, or software instantly through a web console or API, with no human interaction from the provider. In an on-premise setup, provisioning new hardware involves purchase approval, delivery, and manual installation, taking days or weeks.
- **Broad Network Access** — Cloud resources are accessible over the internet from any device or location. On-premise systems are usually confined to an organization's internal network and require VPN access from outside.
- **Resource Pooling & Multi-Tenancy** — A cloud provider's physical infrastructure is shared across many customers using a multi-tenant model, with resources dynamically assigned as needed. On-premise hardware is dedicated entirely to a single organization, regardless of how much of it is actually used.
- **Rapid Elasticity** — Cloud resources can automatically scale up during high demand and scale down afterward. On-premise capacity is fixed by the hardware already purchased, so handling a demand spike requires buying and installing new equipment in advance.

- **Measured, Pay-as-You-Go Service** — Cloud usage is metered like a utility bill, so customers pay only for the resources consumed. On-premise computing requires a large upfront capital investment in hardware that must be paid for whether it is fully used or not.

Q2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.

1. Mainframe Era (1950s–60s)

Large, centralized mainframes served many users at once through time-sharing terminals. This introduced the core idea that later shaped cloud computing: many users sharing one pool of computing power rather than each owning a dedicated machine.

2. Hardware Decentralization (1970s–90s)

The rise of minicomputers and then personal computers moved processing power out of centralized mainframes and onto individual desktops. Client-server architecture then reintroduced some centralization, with dedicated servers handling requests from many client machines over a local network.

3. Virtualization (1990s–2000s)

Hardware virtualization (pioneered commercially by companies such as VMware) allowed one physical server to be divided into multiple independent virtual machines. This decoupled software from physical hardware for the first time, making it possible to share one machine's capacity among many isolated workloads — a critical enabling step for the cloud.

4. Internet and Software Evolution (1990s–2000s)

The growth of the public internet and broadband connectivity made it practical to access applications remotely. Early web-based software — such as Salesforce's 1999 launch of CRM software delivered entirely through a browser — proved that software no longer needed to be installed locally to be useful, laying the groundwork for the SaaS model.

5. Convergence into Modern Cloud Platforms

Modern cloud computing emerged when four threads came together: virtualization (to abstract hardware), high-speed internet (to enable remote access), standardized web-service APIs (to enable self-service automation), and utility-style billing (to enable pay-as-you-go pricing). Amazon Web Services launched in 2006 as the first major public IaaS offering, followed by Google App Engine and Microsoft Azure, marking the arrival of cloud computing as it is known today.

Q3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.

Role of Server Virtualization

Server virtualization uses a hypervisor to abstract a physical server's CPU, memory, and storage into several independent virtual machines, each running its own operating system and applications in isolation. In cloud computing, this is the mechanism that makes several defining features possible:

- **Resource pooling** — many customers' virtual machines can share one physical server safely.
- **Rapid provisioning** — a new virtual machine can be created in minutes, compared with days for a new physical server.
- **Elasticity** — virtual machines can be resized, cloned, or migrated between physical hosts on demand.
- **Higher utilization** — idle physical capacity can be assigned to other workloads instead of sitting unused.

Virtualization vs. Cloud Computing

Virtualization is an enabling TECHNOLOGY — a technical mechanism for partitioning one physical machine into multiple isolated virtual instances. Cloud computing is a SERVICE DELIVERY MODEL — a way of offering computing resources on demand, self-service, over the internet, with metered billing. The two are related but not the same: a company can run VMware inside its own data center purely for internal use, which is virtualization without cloud computing, since it is not self-service, internet-accessible, or metered. Conversely, cloud computing is what happens when a provider takes virtualized (or, increasingly, containerized) resources and wraps them in self-service APIs, internet access, and pay-as-you-go billing. In short, virtualization answers “how are the resources technically abstracted?” while cloud computing answers “how are those resources delivered to and consumed by the customer?”

Q4. Discuss the role of data centers in delivering cloud services.

Data centers are the physical facilities that house the servers, storage systems, networking equipment, and supporting infrastructure — power, cooling, and security — on which every cloud service ultimately runs. They play several essential roles:

- **Physical infrastructure backbone** — thousands of servers organized into racks provide the raw compute, storage, and network capacity that the provider's software then virtualizes and offers to customers.
- **Redundancy and high availability** — redundant power supplies, backup generators, and duplicate network links keep services running through equipment failure; providers also group data centers into multiple availability zones and regions so that the loss of one facility does not cause a total outage.
- **Scalability** — large-scale data centers let a provider add capacity as customer demand grows, which is what makes the cloud's promise of elasticity possible in practice.
- **Security and compliance** — physical safeguards such as biometric access control and continuous surveillance, combined with certifications like ISO 27001 or SOC 2, protect customer data and satisfy regulatory requirements.

- **Global distribution and lower latency** — providers operate data centers across many geographic regions so applications and data can be hosted close to end users, reducing latency and supporting disaster recovery.

In short, data centers are the foundational physical layer of cloud computing — without them, the virtualization and service-delivery layers that customers actually interact with would have nothing to run on.

Q5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

The four service models differ mainly in how much of the technology stack the provider manages versus how much is left to the customer. Moving from IaaS to PaaS to SaaS, the level of abstraction increases and the customer's management responsibility steadily decreases, while the provider takes on more. XaaS extends this same “as-a-service” idea to virtually any IT capability beyond the traditional three.

Model	What It Provides	Customer Manages	Example
IaaS (Infrastructure as a Service)	Virtualized computing infrastructure: virtual machines, storage, and networks.	Operating system, middleware, runtime, and applications.	Amazon EC2
PaaS (Platform as a Service)	A ready-to-use platform — OS, runtime, and development tools — for building and deploying applications.	Only the application code and data.	Google App Engine
SaaS (Software as a Service)	A fully functional, ready-to-use application delivered over the internet, usually through a browser.	Nothing beyond configuration and user data; the provider manages the entire stack.	Google Workspace (Gmail, Docs)
XaaS (Anything/Everything as a Service)	An umbrella term for any IT capability delivered on demand — e.g. Function-, Database-, or Desktop-as-a-Service.	Varies by specific service, but is minimal by design.	AWS Lambda (Function-as-a-Service)

Key distinction: IaaS gives the most control (and the most management responsibility) to the customer, SaaS gives the least, and PaaS sits in between by handling infrastructure while leaving application logic to the developer. XaaS is not a fixed layer but a general pattern — any capability, from storage to security, can be offered “as a service.”